

Summary

Computational physics researcher building distributed scientific software and using machine learning to accelerate calculations for first-principles materials research. Experience spans Python/C++, CUDA/ROCm, MPI/OpenMP, Linux-based HPC environments, and Docker containers for packing applications across environments.

Research Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Implemented INCITE-supported first-principles workflows in Python/C++ with MPI, OpenMP, CUDA/ROCm, PETSc, and SLEPc on Frontier at OLCF and Perlmutter at LBNL.
- Formulated coherent-state and path-integral treatments of exciton-phonon dynamics for self-trapped excitons and exciton-polarons in LiF and lead-free double perovskites, connecting DFT, DFPT, and GW/BSE-style methods.
- Built DeePH-style computational graphs and trained PyTorch Geometric equivariant models with automatic differentiation to accelerate optical-property calculations.
- Packaged scientific applications in Docker containers for deployment across different computing environments and used distributed debuggers such as TotalView and Linaro DDT to track down bugs on HPC systems.
- Co-authored a 2025 *Journal of the American Chemical Society* paper on ultrafast self-trapped exciton formation in lead-free double halide perovskites and presented related APS research in 2024, 2025, and 2026.

Industry Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed CUDA-based software for high-volume acoustic waveform processing, visualization, and field-tool diagnostics in geological well logging systems.
- Built full-stack server applications with C#, ASP.NET, and Microsoft SQL Server for database design, real-time remote tool testing, and calibration of temperature and pressure sensors.
- Applied finite-element analysis with dolfinx for frequency analysis of steel pipes in well environments to support physics-driven engineering studies.

FindLight

Photonics Technical Writer

2018–2019

- Wrote clear technical content on lasers, optics, and photonics devices for engineering audiences and product-focused readers.

Selected Projects

Exciton-Phonon HPC Research: Implemented distributed first-principles calculations for LiF and lead-free double perovskites using CUDA/ROCm, MPI, OpenMP, PETSc, and SLEPc.

Equivariant Scientific ML: Built DeePH-style computational graphs and PyTorch Geometric equivariant models with automatic differentiation for excitonic calculations.

Meep Ring Resonator FDTD: Designed a nanophotonic ring-resonator band-pass filter in telecom bands using Meep-based FDTD simulation.

DOLFINx Finite-Element Analysis: Built 2D DOLFINx/PETSc simulations for transient heat diffusion, Poisson potential around a circular cutout, and incompressible flow past an immersed cylinder, with PyVista and Matplotlib results.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript, Bash, SQL, C#

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, SciPy, SymPy, matplotlib, dolfinx

GPU Libraries: CUDA, ROCm, cuBLAS, cuTENSOR, rocBLAS, rocFFT, CUDA-aware MPI

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks

Platforms: Linux, macOS, Windows, Docker, Docker Compose, Kubernetes, gcloud, Frontier, Perlmutter

Domains: First-principles materials modeling, many-body theory, distributed computing, computational physics, numerical PDEs